CHAPTER 8
Sequences and Progressions

Easy Notes for Class 9

RAVI SHANKAR

1. What is a Sequence?

A **sequence** is a list of numbers arranged in a definite order according to some pattern.

$$2,\ 4,\ 6,\ 8,\ 10, \ldots$$

Here,

- 2 is the first term.
- 4 is the second term.
- 6 is the third term.

The nth term of a sequence is generally written as t_n.

$$\boxed{\text{Sequence} = \text{Numbers} + \text{Order} + \text{Pattern}}$$

2. How to Find the Pattern?

Whenever you see a sequence, first ask:

"How is one term changing into the next term?"

For example,

$$3,\ 6,\ 9,\ 12,\ 15, \ldots$$

Each term increases by 3.
Therefore,

$$15 + 3 = 18$$

So the next term is $\boxed{18}$.

3. Some Important Sequences

Natural Numbers:

$$1,\ 2,\ 3,\ 4,\ 5, \ldots$$

nth term:

$$\boxed{t_n = n}$$

Odd Numbers:

$$1,\ 3,\ 5,\ 7,\ 9, \ldots$$

nth term:

$$\boxed{t_n = 2n - 1}$$

Square Numbers:

$$1,\ 4,\ 9,\ 16,\ 25, \ldots$$

nth term:

$$\boxed{t_n = n^2}$$

4. Triangular Numbers

The triangular numbers are

$$1,\ 3,\ 6,\ 10,\ 15,\ 21, \ldots$$

Notice:

$$\begin{aligned} 1 &= 1 \\ 3 &= 1 + 2 \\ 6 &= 1 + 2 + 3 \\ 10 &= 1 + 2 + 3 + 4 \end{aligned}$$

Therefore, the nth triangular number is

$$\boxed{t_n = \frac{n(n+1)}{2}}$$

5. Explicit Rule

An **explicit rule** gives us a direct formula for finding any term of a sequence.
For example,

$$3,\ 5,\ 7,\ 9,\ 11, \ldots$$

We can observe:

$$\begin{aligned} 3 &= 2(1) + 1 \\ 5 &= 2(2) + 1 \\ 7 &= 2(3) + 1 \end{aligned}$$

Therefore,

$$\boxed{t_n = 2n + 1}$$

For example, the 20th term is

$$t_{20} = 2(20) + 1 = 41$$

6. Recursive Rule

A **recursive rule** finds the next term using the previous term.
Consider:

$$2,\ 5,\ 8,\ 11,\ 14, \ldots$$

Every term is obtained by adding 3 to the previous term.
Therefore,

$$\boxed{t_{n+1} = t_n + 3}$$

with

$$\boxed{t_1 = 2}$$

Recursive Rule: Previous Term $\longrightarrow$ Next Term

7. Fibonacci-Type Sequence

In this type of sequence, each term is obtained by adding the previous two terms.
For example,

$$1,\ 2,\ 3,\ 5,\ 8,\ 13,\ 21, \ldots$$

because

$$\begin{aligned} 1 + 2 &= 3 \\ 2 + 3 &= 5 \\ 3 + 5 &= 8 \\ 5 + 8 &= 13 \end{aligned}$$

The rule is

$$\boxed{t_n = t_{n-1} + t_{n-2}}$$

8. Arithmetic Progression (AP)

A sequence is called an **Arithmetic Progression (AP)** if the difference between consecutive terms is always the same.
Example:

$$5,\ 8,\ 11,\ 14,\ 17, \ldots$$

Here,

$$\begin{aligned} 8 - 5 &= 3 \\ 11 - 8 &= 3 \\ 14 - 11 &= 3 \end{aligned}$$

Thus, the common difference is

$$\boxed{d = 3}$$

Important AP Formula

If the first term is a and the common difference is d, then

$$\boxed{t_n = a + (n-1)d}$$

where

- a = first term
- d = common difference
- n = position of the term
- t_n = nth term

Example: Find the 10th term

Consider the AP

$$4,\ 7,\ 10,\ 13, \dots$$

Here,

$$a = 4, \qquad d = 3$$

Using

$$t_n = a + (n-1)d$$

we get

$$\begin{aligned} t_{10} &= 4 + (10-1)(3) \\ &= 4 + 27 \\ &= \boxed{31} \end{aligned}$$

9. Geometric Progression (GP)

A sequence is called a **Geometric Progression (GP)** if the ratio between consecutive terms is always the same.
Example:

$$2,\ 6,\ 18,\ 54, \ldots$$

Here,

$$\frac{6}{2} = 3$$
$$\frac{18}{6} = 3$$
$$\frac{54}{18} = 3$$

Therefore, the common ratio is

$$\boxed{r = 3}$$

Important GP Formula

If the first term is a and the common ratio is r, then

$$\boxed{t_n = ar^{n-1}}$$

Example: Find the 5th term

Consider

$$3,\ 6,\ 12,\ 24, \dots$$

Here,

$$a = 3, \qquad r = 2$$

Therefore,

$$\begin{aligned} t_5 &= ar^{n-1} \\ &= 3(2)^{5-1} \\ &= 3(2)^4 \\ &= 48 \end{aligned}$$

Hence,

$$\boxed{t_5 = 48}$$

10. AP vs GP — Very Easy Trick

AP	**GP**	**Example**
Same Difference	Same Ratio	—
Add/Subtract	Multiply/Divide	—
$2, 5, 8, 11$	$2, 6, 18, 54$	—

$$\boxed{\text{AP} \to \text{Same Difference}}$$

$$\boxed{\text{GP} \to \text{Same Ratio}}$$

Sum of First n Natural Numbers

The sum

$$1 + 2 + 3 + \cdots + n$$

is

$$\boxed{S_n = \frac{n(n+1)}{2}}$$

For example, the sum of the first 10 natural numbers is

$$\begin{aligned} S_{10} &= \frac{10(10+1)}{2} \\ &= \frac{10 \times 11}{2} \\ &= \boxed{55} \end{aligned}$$

One-Minute Revision

Remember these ideas:

$$\boxed{\text{Sequence} = \text{Pattern of Terms}}$$

$$\boxed{\text{Explicit Rule} = \text{Directly Find the Term}}$$

$$\boxed{\text{Recursive Rule} = \text{Previous Term} \to \text{Next Term}}$$

$$\boxed{\text{AP} = \text{Same Difference}}$$

$$\boxed{t_n = a + (n-1)d}$$

$$\boxed{\text{GP} = \text{Same Ratio}}$$

$$\boxed{t_n = ar^{n-1}}$$

$$\boxed{\text{Triangular Number} = \frac{n(n+1)}{2}}$$

Think → Find the Pattern → Apply the Rule → Solve!

Mathematics becomes easy when the pattern becomes clear.